C# PRACTICAL 3

Delegates, Multicast Delegate and BankAccount Property Validation

Practical 3a: Arithmetic Methods via C# Delegates

Code:

```
using System;

namespace caldelegate
{
    delegate double ArithmeticOperation(double a, double b);

    class Calculator
    {
        public static double Add(double a, double b)
        {
            return a + b;
        }

        public static double Subtract(double a, double b)
        {
            return a - b;
        }

        public static double Multiply(double a, double b)
        {
            return a * b;
        }

        public static double Divide(double a, double b)
        {
            if (b == 0)
            {
                Console.WriteLine("Division by zero is not possible.");
                return 0;
            }
            return a / b;
        }

        static void Main(string[] args)
        {
            Console.Write("Enter First Number: ");
            double num1 = Convert.ToDouble(Console.ReadLine());

            Console.Write("Enter Second Number: ");
            double num2 = Convert.ToDouble(Console.ReadLine());

            ArithmeticOperation op;

            op = Add;
            Console.WriteLine("Addition = " + op(num1, num2));

            op = Subtract;
            Console.WriteLine("Subtraction = " + op(num1, num2));

            op = Multiply;
            Console.WriteLine("Multiplication = " + op(num1, num2));

            op = Divide;
            Console.WriteLine("Division = " + op(num1, num2));
        }
    }
}
```

Output:

```
Enter First Number: 300
Enter Second Number: 200
Addition = 500
Subtraction = 100
Multiplication = 60000
Division = 1.5
```

Practical 3b: Multicast Delegate

Code:

```
using System;

namespace multicast_delegate
{
    delegate void Mdelegate();

    class DM
    {
        public static void display()
        {
            Console.WriteLine("hello");
        }

        public static void Print()
        {
            Console.WriteLine("hello world");
        }
    }

    class Mtest
    {
        public static void Main()
        {
            Mdelegate m1 = new Mdelegate(DM.display);
            Mdelegate m2 = new Mdelegate(DM.Print);
            Mdelegate m3 = m1 + m2;
            Mdelegate m4 = m2 + m1;
            Mdelegate m5 = m3 - m2;

            Console.WriteLine("delegate M3:");
            m3();

            Console.WriteLine("delegate M4:");
            m4();

            Console.WriteLine("delegate M5:");
            m5();
        }
    }
}
```

Output:

```
Multicast delegate
delegate M3:
hello
hello world
delegate M4:
hello world
hello
delegate M5:
hello
```

Practical 3c: BankAccount Class with Property Validation

Code:

```
using System;

namespace bankaccount
{
    class BankAccount
    {
        // Private field for account number
        private string accountnumber;

        // Property for AccountNumber
        public string AccountNumber
        {
            get { return accountnumber; }
            set { accountnumber = value; }
        }

        // Private field for balance
        private double balance;

        // Property with validation
        public double Balance
        {
            get { return balance; }
            set
            {
                if (value >= 0)
                {
                    balance = value;
                }
                else
                {
                    Console.WriteLine("Balance cannot be negative.");
                }
            }
        }

        // Constructor
        public BankAccount(string accountNumber, double balance)
        {
            AccountNumber = accountNumber;
            Balance = balance;
        }

        // Method to display account details
        public void Display()
        {
            Console.WriteLine("\nAccount Details");
            Console.WriteLine("Account Number: " + AccountNumber);
            Console.WriteLine("Balance: " + Balance);
        }
    }

    class Program
    {
        static void Main()
        {
            Console.Write("Enter Account Number: ");
            string accNo = Console.ReadLine();

            Console.Write("Enter Balance: ");
            double bal = Convert.ToDouble(Console.ReadLine());

            BankAccount account = new BankAccount(accNo, bal);
            account.Display();

            Console.ReadKey();
        }
    }
}
```

Output:

```
Enter Account Number: 123456789
Enter Balance: 5000

Account Details
```

```
Account Number: 123456789
Balance: 5000
```